Skeletal Muscle Growth in Fish: An Integrated Review

General Background

Fish muscle growth is not a single process: it reflects the balance between enlargement of existing fibers, or hypertrophy, and the recruitment of new fibers, or hyperplasia (Johnston et al., 2011). pubmed.ncbi.nlm.nih.gov This distinction provides the basis for interpreting growth studies, because an increase in body or muscle mass alone cannot show which cellular process produced it (Rowlerson et al., 1995). In many teleosts, muscle precursor cells remain available beyond embryonic development, which allows fiber recruitment to continue during juvenile and, in some species, adult growth (Manneken et al., 2022). pubmed.ncbi.nlm.nih.gov The resulting developmental flexibility is supported by a myogenic program involving cell proliferation, migration, fusion, and differentiation, although the genes involved may have species-specific duplicates and functions (Rossi & Messina, 2014). link.springer.com Thus, muscle growth in fish is best treated as a changing cellular outcome shaped by species, developmental stage, and environment, rather than as a fixed progression that applies uniformly across teleosts (Perez et al., 2023). pubmed.ncbi.nlm.nih.gov

This framework also explains why growth must be measured at more than one biological level. Histology can show changes in fiber number and size, while gene-expression studies can identify candidate regulatory processes; neither measure alone answers every question about muscle accretion (Johnston et al., 2011). pubmed.ncbi.nlm.nih.gov The distinction is particularly relevant because fish muscle contains evolutionarily diverse sarcomeric proteins, and similar gene names across species do not always imply identical functions (Joyce, 2023). pure.au.dk

Developmental Changes and the Cellular Basis of Growth

The central developmental question is not simply whether fish muscle grows through hyperplasia or hypertrophy, but how their relative contributions change over time. Early morphometric work in gilthead sea bream showed that both processes occur, with hyperplasia especially prominent during larval development and hypertrophy contributing across life stages (Rowlerson et al., 1995). A recent, long-term rainbow trout trial complicates a simple transition from early hyperplasia to later hypertrophy: across fish from 2.2 to 350 g, the authors identified successive phases in which hyperplasia and hypertrophy alternated (Singha et al., 2025). pubmed.ncbi.nlm.nih.gov This pattern means that body size and sampling time affect how a study characterizes growth, and that findings from one species or stage should not be treated as a general developmental rule (Singha et al., 2025). www.cambridge.org

Other approaches show why the cellular pattern can be difficult to infer from molecular data alone. In pikeperch, expression of IGF-family genes and

myogenic regulators varied across ontogeny, indicating stage-specific changes in the growth program without directly establishing when new fibers were added (Lavajoo Bolgouri et al., 2024). pmc.ncbi.nlm.nih.gov Spatial transcriptomics in zebrafish adds a complementary finding: adult muscle is regionally heterogeneous, and some areas retain gene-expression patterns associated with active myogenesis and hyperplasia (Liu et al., 2022). pubmed.ncbi.nlm.nih.gov Together, these studies make a strong case for stage- and region-specific sampling, while also showing why transcript profiles should be interpreted as evidence about candidate processes rather than as direct measurements of fiber growth (Liu et al., 2022). pubmed.ncbi.nlm.nih.gov

Molecular Regulation and Nutritional Effects

At the molecular level, the literature describes interactions among myogenic regulators, growth-factor signaling, protein synthesis, and protein degradation rather than a single pathway acting independently (Duran et al., 2022). www.mdpi.com In pacu myotubes, transcriptomic and microRNA analyses after amino-acid or IGF-1 stimulation identified candidate networks involved in this response, but the cell-culture design cannot establish how strongly those networks determine whole-fish growth (Duran et al., 2022). pmc.ncbi.nlm.nih.gov A related gilthead sea bream study broadened the molecular picture by examining changes in microRNAs and long non-coding RNAs after amino-acid and IGF-1 stimulation of myoblasts (García-Pérez et al., 2024). These studies increase the range of candidate regulators, but their value for explaining muscle accretion depends on follow-up evidence at the level of fibers and whole-animal growth (García-Pérez et al., 2024).

Nutritional experiments provide a bridge between those candidate pathways and measurable phenotypes, although their outcomes depend on which nutrient and endpoint are studied. In juvenile turbot, methionine deficiency reduced growth and muscle protein while changing markers associated with myogenesis, mTOR signaling, and protein metabolism (Sui et al., 2023). www.sciencedirect.com In hybrid catfish, dietary tryptophan altered muscle protein content and fiber characteristics alongside protein-synthesis and myogenic markers, linking molecular changes with tissue-level outcomes in the same feeding experiment (Jiang et al., 2022). pubmed.ncbi.nlm.nih.gov Common-carp research on branched-chain amino acids has used transcriptome and microRNA sequencing to map candidate regulatory pathways, but those data do not by themselves show whether fiber recruitment or fiber size changed (Cao et al., 2023). These differences in study design matter: a change in gene expression, a change in fiber morphology, and an increase in muscle mass are related outcomes, but they are not interchangeable evidence.

Several recent feeding trials have measured muscle growth more directly. In juvenile largemouth bass, dietary creatine was associated with greater

protein deposition and myofiber hyperplasia (Yu et al., 2024). In adult grass carp, dietary myo-inositol was investigated as a means of promoting hypertrophic growth, extending nutritional research to a later life stage (Wang et al., 2025). pubmed.ncbi.nlm.nih.gov The rainbow trout trial further showed that alternative plant-protein diets and additive mixtures can affect fiber recruitment differently across growth phases, reinforcing the need to evaluate diet in relation to developmental stage (Singha et al., 2025). pubmed.ncbi.nlm.nih.gov In gibel carp, researchers tested whether essential-amino-acid supplementation could maintain growth-related outcomes when plant proteins replaced fishmeal, illustrating how muscle-growth research also addresses changes in aquaculture feed composition (Cai et al., 2021). A micronutrient intervention adds another dimension: selenium deficiency in juvenile zebrafish was associated with restricted fiber hypertrophy and lower TORC1-mediated protein synthesis (Wang et al., 2023). pubmed.ncbi.nlm.nih.gov Collectively, these feeding studies support nutrient-sensitive muscle growth, but they do not point to one universal dietary solution across species or developmental stages (Perez et al., 2023). pubmed.ncbi.nlm.nih.gov

Environmental Conditions, Activity, and Recovery from Fasting

Environmental conditions matter both because they influence growth and because they complicate comparisons among experiments. Temperature experiments in spotted sea bass primary muscle cells measured proliferation, differentiation, and migration under different thermal conditions, clarifying cellular responses while leaving whole-animal growth to be tested separately (Zhang et al., 2022). www.mdpi.com In Nile tilapia, embryonic temperature was associated with later differences in muscle circular-RNA expression and somatic growth, suggesting a persistent developmental association without establishing that individual RNA changes caused the growth effect (Rbbani et al., 2024). www.frontiersin.org Field observations of an intertidal fish have also linked coastal upwelling conditions with muscle histology and growth-related molecular markers, bringing natural environmental variation into a literature dominated by controlled feeding and cell-culture studies (Zuloaga et al., 2023). pubmed.ncbi.nlm.nih.gov

Exercise research illustrates the same distinction between a measurable muscle phenotype and the pathway proposed to explain it. In adult zebrafish, swimming training was associated with larger fast-muscle fibers and greater vascularization, together with changes in myogenic and angiogenic transcriptional programs (Palstra et al., 2014). research.monash.edu In juvenile sea bream, exercise and diet were tested together, and the hormonal and myogenic responses varied with the feeding and swimming conditions (Perelló-Amorós et al., 2021). pubmed.ncbi.nlm.nih.gov These results make it difficult to treat exercise as a uniform growth stimulus: its

effect depends on the species, exercise regime, diet, and outcome measured (Canosa & Bertucci, 2023). pubmed.ncbi.nlm.nih.gov

Fasting and refeeding provide a useful model for examining how these processes respond to changing conditions, but “compensatory growth” can refer to recovery of body mass, muscle mass, or a previous growth trajectory (Ali et al., 2022). www.mdpi.com In pejerrey, measurements of muscle growth and cell proliferation during fasting and refeeding showed that hyperplasia and hypertrophy responded differently over the course of the experiment (Simó et al., 2022). pubmed.ncbi.nlm.nih.gov A meta-analysis of fish muscle responses to fasting and refeeding identified recurring changes in genes associated with protein synthesis and cellular regulation, while also bringing together studies that differ in species and treatment design (Perez et al., 2022). pubmed.ncbi.nlm.nih.gov The synthesis therefore supports fasting and refeeding as a way to investigate muscle plasticity, while cautioning against treating all forms of post-fasting recovery as the same biological outcome (Perez et al., 2022).

Myostatin and Causal Evidence

Research on myostatin illustrates the difference between identifying a candidate regulator and testing its function. In Nile tilapia, antisense-mediated myostatin knockdown was assessed using muscle histology and tissue-composition measures, allowing the study to examine muscle outcomes in addition to transcript change (Wang et al., 2023). pubmed.ncbi.nlm.nih.gov CRISPR/Cas9 disruption of *mstn* in large yellow croaker produced mosaic mutants, establishing the feasibility of gene editing in that species but not, on its own, a settled muscle-growth phenotype (Yan et al., 2022). pubmed.ncbi.nlm.nih.gov In channel catfish, myostatin knockout was evaluated against growth and disease-resistance outcomes, moving the work toward traits relevant to production (Coogan et al., 2022). www.sciencedirect.com A 2026 grass-carp study reported increased muscle hypertrophy following *mstnb* knockout, extending the evidence to another species and a directly measured fiber phenotype (Sun et al., 2026). www.sciencedirect.com These results strengthen the case that myostatin can regulate fish muscle growth, but they do not establish identical effects across species or show that greater muscle mass necessarily comes with the same changes in fiber number, body shape, or other production traits (Sun et al., 2026). www.sciencedirect.com

Synthesis

The evidence is strongest for the broad conclusion that the balance between fiber recruitment and fiber enlargement changes with developmental stage and can respond to nutrition, activity, and environmental conditions (Singha et al., 2025). pubmed.ncbi.nlm.nih.gov It is less conclusive about how specific molecular signals cause those changes, because expression profiles

and cell-culture responses do not establish fiber growth in intact animals (Lavajoo Bolgouri et al., 2024). pubmed.ncbi.nlm.nih.gov This difference in evidential strength suggests a clear priority for future research: combine repeated histology with protein-turnover or cell-proliferation measurements and functional perturbations, so that molecular changes can be connected to a defined cellular outcome (Simó et al., 2022). pubmed.ncbi.nlm.nih.gov

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